

Unggi Lee

Real-time systems · DNN inference offloading · Image-adaptive compression

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EDUCATION

Daegu Gyeongbuk Institute of Science and Technology (DGIST) <i>M.S. in Computer Science</i>	Feb. 2026 - Feb. 2028 (Expected) <i>Daegu, S.Korea</i>
Real-Time Computing Lab (RTCL)	
Daegu Gyeongbuk Institute of Science and Technology (DGIST) <i>B.S. in Electrical Engineering and Computer Science</i>	Feb. 2020 - Feb. 2026 <i>Daegu, S.Korea</i>
Total GPA of 3.82 / 4.30	

RESEARCH EXPERIENCE

Real-Time Computing Lab (RTCL), DGIST <i>Graduate Research Assistant</i>	Feb. 2026 - Present <i>Daegu, S.Korea</i>
- Real-time systems, with a focus on DNN inference offloading between edge devices and servers under latency constraints. - Image-adaptive JPEG compression: per-image quality factor and quantization table optimization, so that offloaded inference holds up under variable network bandwidth.	

PROJECT EXPERIENCE

WardrobeRL: Prompt-Aware Reinforcement Learning Based Top-Bottom Recommendation <i>Reinforcement Learning course project, published at ICTC 2025</i>	2025 <i>Daegu, S.Korea</i>
A model-free PPO framework that assembles top-bottom outfit pairs from a natural-language style prompt, rewarded by CLIP similarity fused with a learned proxy for human preference.	
Noise Map-Driven Dynamic Frame Processing for Improved Object Detection in Low-Light Conditions <i>Undergraduate Group Research Program (UGRP), Team of Four</i>	2024 <i>Daegu, S.Korea</i>
- Placed an enhancement front-end before YOLO – Zero-DCE for low-light enhancement, CBDNet for noise-map estimation, and MFDNet for denoising – integrated directly into the detector's data loader. - Held the pipeline to real time by exploiting temporal redundancy between frames, processing one frame in five at 20 FPS. - Improved mAP by roughly 1% over the unenhanced baseline on BDD100k.	

PUBLICATIONS

WardrobeRL: Prompt-Aware Reinforcement Learning Based Top-Bottom Recommendation <i>Minseok Oh, Unggi Lee, Hyunmin Kim, Dooseek Lee</i>	Oct. 2025 <i>ICTC 2025</i>
2025 16th International Conference on Information and Communication Technology Convergence (ICTC) – PDF / IEEE Xplore	

INTERNATIONAL EXPERIENCE

University of California, Los Angeles (UCLA) <i>Summer Session</i>	Jun. - Aug. 2023 <i>Los Angeles, CA, USA</i>
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SKILLS

Programming	C, C++, Python
Deep Learning	PyTorch
Computer Vision	OpenCV
Tools	Docker, Linux, Git